EXTENSION BY THE EMPTY SET IS LEFT ADJOINT TO RESTRICTION, WITH STALKS AND COUNIT IDENTITIES

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Allowed standard prerequisites (editorial).

- The restriction description of j_p: maps F -> j_pG are compatible families F(V) -> G(V) for V open in U (lemma-j-pullback). invoked at proof lines 4802-4805. Identifies maps out of j_{p!}F with maps into the restriction, proving (1).

Source reference. Original lemma 00A0: sources/stacks-sheaves/pinned-original-source.tex, source lines 4669–4669. Cited at proof lines 4802-4805 to identify compatible families with maps F -> j_pG; load-bearing for part (1) inside the counted span.

Lemma 0.2. *Let X be a topological space. Let $j : U \to X$ be the inclusion of an open subset.*

(1) *The functor $j_{p!}$ is a left adjoint to the restriction functor j_p (see Lemma 00A0).*
(2) *The functor $j_!$ is a left adjoint to restriction, in a formula*
$$\mathrm{Mor}_{Sh(X)}(j_!\mathcal{F}, \mathcal{G}) = \mathrm{Mor}_{Sh(U)}(\mathcal{F}, j^{-1}\mathcal{G}) = \mathrm{Mor}_{Sh(U)}(\mathcal{F}, \mathcal{G}|_U)$$
bifunctorially in $\mathcal{F}$ and $\mathcal{G}$.
(3) *Let $\mathcal{F}$ be a sheaf of sets on U. The stalks of the sheaf $j_!\mathcal{F}$ are described as follows*
$$j_!\mathcal{F}_x = \begin{cases} \emptyset & \text{if} \quad x \notin U \\ \mathcal{F}_x & \text{if} \quad x \in U \end{cases}$$
(4) *On the category of presheaves of U we have $j_p j_{p!} = id$.*
(5) *On the category of sheaves of U we have $j^{-1} j_! = id$.*

Proof. To map $j_{p!}\mathcal{F}$ into $\mathcal{G}$ it is enough to map $\mathcal{F}(V) \to \mathcal{G}(V)$ whenever $V \subset U$ compatibly with restriction mappings. And by Lemma 00A0 the same description holds for maps $\mathcal{F} \to \mathcal{G}|_U$. The adjointness of $j_!$ and restriction follows from this and the properties of sheafification. The identification of stalks is obvious from the definition of the extension by the empty set and the definition of a stalk. Statements (4) and (5) follow by computing the value of the sheaf on any open of U. □

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